Seminar 5 - Lectures 10 and 11
Computer Vision 1, Master AI, 2026

Exercises inherited from Arnoud Visser, Martin Oswald, Qi Bi, Roan van Blanken

1 Exercise 1: Convolutional Layer Arithmetics

In this exercise, you will learn the relationships between fully connected (FC) and convolution layers. It is important to understand the advantages and downsides of these two fundamental layers in neural networks.

Q.1.a Given an image I with dimensions $H \times W \times C$ as $100 \times 100 \times 1$, a fully connected (FC) layer, $\mathcal{F} : (H \times W \times C) \rightarrow D$, is used to map the *whole* image (I) into a feature vector $v \in \mathbb{R}^D$ with $D = 1000$. How many parameters are in the FC layer (bias is included)?

Q.1.b Since using $I \in \mathbb{R}^{100\times100\times1}$ as the input of a FC layer is computationally expensive, we can divide the image I into patches of 10×10 pixels, and then apply the same FC layer to each patch. The FC layer outputs a feature vector $v \in \mathbb{R}^D$ for each of its inputs, where $D = 1000$. Calculate the number of parameters in a FC layer. Assuming that there are no overlapping patches, how many feature vectors are produced from the FC layer?

Q.1.c Describe the setting of a convolutional layer (kernel size, stride, padding) which produces the same output as the FC layer described in **Q.1.a** when applied to image I.

Q.1.d Describe the setting of a convolution layer (kernel size, stride, padding) which produces the same output as the FC layer described in **Q.1.b** when applied to the image I.

Now that you have an understanding of the differences and similarities between the fully connected (FC) layer and the convolutional (CONV) layer, you might want to explore how these two types of layers can be connected.[1]

Next, let's move on to the setting of a convolutional layer, as shown in Figure 1.

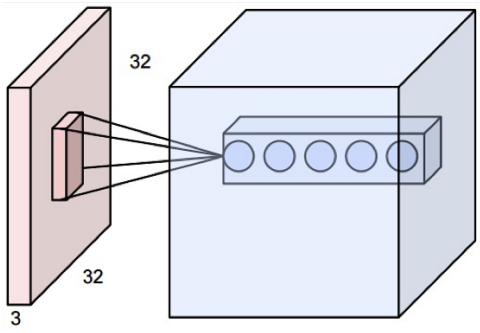


Figure 1: Input size: $32 \times 32 \times 3$, receptive field: 5×5, number of output features: 5 along the depth dimension

Q.1.e What is the size of the output volume with stride 1? (Without padding)

Q.1.f How many weights are learned in one filter? How many in this layer?

Q.1.g What is the depth of a filter in the next layer?

Q.1.h What is the size of the output volume with stride 3?

Q.1.i How many weights are learned with stride 3?

[1] For more details on the connection between FC and CONV layers, refer to the **Converting FC layers to CONV layers** section in this comprehensive note.

2 Exercise 2: Weight Sharing

Q.2.a What are the properties and/or advantages of weight sharing in a convolutional layer?

Q.2.b Describe a situation where weight sharing is not beneficial for recognition.

3 Exercise 3: Gradient

The gradient is an important factor for the successful training of neural networks (read this article if you are not convinced). This part will walk you through some common problems in training convolutional neural networks.

Q.3.a Given an input matrix and a convolutional layer with a weight matrix (kernel):

$$x = \begin{bmatrix} 0 & -1 & 3 \\ 2 & 0.5 & 1 \end{bmatrix} \in \mathbb{R}^{2\times 3\times 1}, \quad W = \begin{bmatrix} 0 & 1 \\ 1 & 0.5 \end{bmatrix} \in \mathbb{R}^{2\times 2\times 1}$$

and no bias. Assume stride = 1 and padding = 0. Compute the output matrix $o \in \mathbb{R}^{1\times 2\times 1}$.

Q.3.b What is the gradient with respect to the input matrix x, given that the gradient with respect to the output matrix o is

$$\begin{bmatrix} 1 & 1 \end{bmatrix}$$

Q.3.c Consider the ReLU non-linearity $z = \max(0, x)$, where the input matrix is $x = \begin{bmatrix} 0 & -1 \\ 2 & 0.5 \end{bmatrix} \in \mathbb{R}^{2\times 2}$. First, calculate the matrix z. Then, given that the gradient w.r.t. z is $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, find the gradient w.r.t. x.

Q.3.d Read this post on the *Dying* ReLU problem here. Describe and explain possible ways to prevent this problem in training convolutional neural networks.

Q.3.e A common problem in training deep neural networks is gradient vanishing and exploding, as discussed in this article. List some common solutions for this problem and explain them.

4 Exercise 4: Single Shot Object Detection

Q.4.a What are the main differences between Single Shot Detectors like YOLO and older two-stage object detection methods like R-CNN?

Read sections 1 (Introduction) and 4 (Related Work) of the YOLO v4 paper. The purpose of the questions below is to explore the variety of methods recently developed in object detection.

Q.4.b Mention two backbone networks that we discussed in class.

Q.4.c We have discussed feature pyramids in class. How do the authors of the YOLO v4 paper categorize these types of architectural features?

Q.4.d In paragraph 2.2 (Bag of Freebies) of the paper, the authors describe a problem with using box coordinates in the output objective function. What is this problem, and how can it be solved?

5 Exercise 5: 3D Convolutions

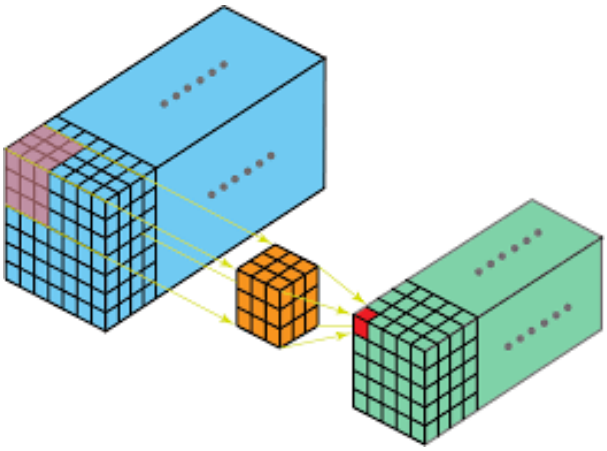


Figure 2: Demonstration of 3D convolution.

Consider a grayscale video with 16 frames, each of size 224×224. You want to generate a feature map with 64 output channels using $3 \times 3 \times 3$ convolutional filters. Proper padding is applied.

Q.5.a What is the dimension of the output weight tensor of the 3D layer? How many parameters are there in total?

Q.5.b What is the dimensionality of the output feature maps if stride = 1 for all 3 dimensions?

Q.5.c What is the dimensionality of the output feature maps if stride = 2 for the temporal dimension and stride = 1 for the spatial dimensions?

Q.5.d Suppose that a standard 3D convolution layer takes an input feature matrix F of shape (l_F, w_F, h_F, c_F) and yields a feature matrix G of size (l_G, w_G, h_G, c_G), where c denotes the number of channels. Also suppose the kernel size is $k \times k \times k$. What is the computational cost of a 3D convolution here?

Q.5.e We can compare a 3D convolution with a 2D convolution over each of the video frames. What is an advantage of such (2+1)D convolutions over 3D convolutions?